```
Initial observations:
--- Climate Records ---
[10:00 | Mumbai | 36.0 C] -> [11:00 | Mumbai | 38.0 C] -> [12:00 | Mumbai | 40.0 C]
-> [13:00 | Mumbai | 42.0 C] -> NULL

Inserting new observation (13:30, Mumbai, 43.0 C):

--- Climate Records ---
[10:00 | Mumbai | 36.0 C] -> [11:00 | Mumbai | 38.0 C] -> [12:00 | Mumbai | 40.0 C]
-> [13:00 | Mumbai | 42.0 C] -> [13:30 | Mumbai | 43.0 C] -> NULL

Searching for 12:00:
Record Found: [12:00 | Mumbai | 40.0 C]

Updating 11:00 temperature:
Temperature at 11:00 updated to 39.5 C

--- Climate Records ---
[10:00 | Mumbai | 36.0 C] -> [11:00 | Mumbai | 39.5 C] -> [12:00 | Mumbai | 40.0 C]
-> [13:00 | Mumbai | 42.0 C] -> [13:30 | Mumbai | 43.0 C] -> NULL
```

Output

```
[10:00 | Mumbai | 36.0 C] -> [11:00 | Mumbai | 38.0 C] -> [12:00 | Mumbai | 40.0 C]
-> [13:00 | Mumbai | 42.0 C] -> [13:30 | Mumbai | 43.0 C] -> NULL

Searching for 12:00:
Record Found: [12:00 | Mumbai | 40.0 C]

Updating 11:00 temperature:
Temperature at 11:00 updated to 39.5 C

--- Climate Records ---
[10:00 | Mumbai | 36.0 C] -> [11:00 | Mumbai | 39.5 C] -> [12:00 | Mumbai | 40.0 C]
-> [13:00 | Mumbai | 42.0 C] -> [13:30 | Mumbai | 43.0 C] -> NULL

Deleting 10:00 record:
Record at 10:00 deleted successfully.

--- Climate Records ---
[11:00 | Mumbai | 39.5 C] -> [12:00 | Mumbai | 40.0 C] -> [13:00 | Mumbai | 42.0 C]
-> [13:30 | Mumbai | 43.0 C] -> NULL
```

```
#include<stdio.h>
#include<stdlib.h>
#include<string.h>

struct WeatherNode
{
   char timestamp[10];
   char location[50];
   float temperature;
   struct WeatherNode *next;
};

struct WeatherNode* createNode(char time[],char loc[],float temp)
{
   struct WeatherNode *newNode;
   newNode=(struct WeatherNode*)malloc(sizeof(struct WeatherNode));
   strcpy(newNode->timestamp,time);
   strcpy(newNode->location,loc);
   newNode->temperature=temp;
   newNode->next=NULL;
   return newNode;
}

void insert(struct WeatherNode **head,char time[],char loc[],float temp)
{
   struct WeatherNode *newNode,*p;
   newNode=createNode(time,loc,temp);
   if(*head==NULL)
      *head=newNode;
   else
   {
      p=*head;
      while(p->next!=NULL)
         p=p->next;
      p->next=newNode;
   }
}

void display(struct WeatherNode *head)
{
   struct WeatherNode *p=head;
   if(p==NULL)
   {
      printf("No climate records available.\n");
```

```
        return;
    }
    printf("\n--- Climate Records ---\n");
    while(p!=NULL)
    {
        printf("[%s | %s | %.1f C]",p->timestamp,p->location,p->temperature);
        if(p->next!=NULL)
            printf(" -> ");
        p=p->next;
    }
    printf(" -> NULL\n");
}

void search(struct WeatherNode *head,char time[])
{
    struct WeatherNode *p=head;
    while(p!=NULL)
    {
        if(strcmp(p->timestamp,time)==0)
        {
            printf("Record Found: [%s | %s | %.1f C]\n",p->timestamp,p->location,p->temperature);
            return;
        }
        p=p->next;
    }
    printf("Record not found.\n");
}

void update(struct WeatherNode *head,char time[],float temp)
{
    struct WeatherNode *p=head;
    while(p!=NULL)
    {
        if(strcmp(p->timestamp,time)==0)
        {
            p->temperature=temp;
            printf("Temperature at %s updated to %.1f C\n",time,temp);
            return;
        }
        p=p->next;
    }
    printf("Record not found.\n");
}
```

```
void delete(struct WeatherNode **head,char time[])
{
   struct WeatherNode *p=*head,*q=NULL;
   while(p!=NULL&&strcmp(p->timestamp,time)!=0)
   {
      q=p;
      p=p->next;
   }
   if(p==NULL)
   {
      printf("Record not found.\n");
      return;
   }
   if(q==NULL)
      *head=p->next;
   else
      q->next=p->next;
   free(p);
   printf("Record at %s deleted successfully.\n",time);
}

int main()
{
   struct WeatherNode *head=NULL;

   insert(&head,"10:00","Mumbai",36.0);
   insert(&head,"11:00","Mumbai",38.0);
   insert(&head,"12:00","Mumbai",40.0);
   insert(&head,"13:00","Mumbai",42.0);

   printf("Initial observations:");
   display(head);

   printf("\nInserting new observation (13:30, Mumbai, 43.0 C):\n");
   insert(&head,"13:30","Mumbai",43.0);
   display(head);

   printf("\nSearching for 12:00:\n");
   search(head,"12:00");

   printf("\nUpdating 11:00 temperature:\n");
   update(head,"11:00",39.5);
   display(head);
```

```
    printf("\nDeleting 10:00 record:\n");
    delete(&head,"10:00");
    display(head);

    return 0;
}
```

main.c

Run ▶

```c
#include<stdio.h>
#include<stdlib.h>
#include<string.h>

struct WeatherNode
{
    char timestamp[10];
    char location[50];
    float temperature;
    struct WeatherNode *next;
};

struct WeatherNode* createNode(char time[],char loc[],float temp)
{
    struct WeatherNode *newNode;
    newNode=(struct WeatherNode*)malloc(sizeof(struct WeatherNode));
    strcpy(newNode->timestamp,time);
    strcpy(newNode->location,loc);
    newNode->temperature=temp;
    newNode->next=NULL;
    return newNode;
```

main.c

Run ▶

```
}

void insert(struct WeatherNode **head,char time[],char loc[],float temp)
{
    struct WeatherNode *newNode,*p;
    newNode=createNode(time,loc,temp);
    if(*head==NULL)
        *head=newNode;
    else
    {
        p=*head;
        while(p->next!=NULL)
            p=p->next;
        p->next=newNode;
    }
}

void display(struct WeatherNode *head)
{
    struct WeatherNode *p=head;
    if(p==NULL)
```

main.c

Run ▶

```
    {
        printf("No climate records available.\n");
        return;
    }
    printf("\n--- Climate Records ---\n");
    while(p!=NULL)
    {
        printf("[%s | %s | %.1f C]",p->timestamp,p->location,p->temperature);
        if(p->next!=NULL)
            printf(" -> ");
        p=p->next;
    }
    printf(" -> NULL\n");
}

void search(struct WeatherNode *head,char time[])
{
    struct WeatherNode *p=head;
    while(p!=NULL)
    {
        if(strcmp(p->timestamp,time)==0)
```

main.c

Run ▶

```c
int main()
{
    struct WeatherNode *head=NULL;

    insert(&head,"10:00","Mumbai",36.0);
    insert(&head,"11:00","Mumbai",38.0);
    insert(&head,"12:00","Mumbai",40.0);
    insert(&head,"13:00","Mumbai",42.0);

    printf("Initial observations:");
    display(head);

    printf("\nInserting new observation (13:30, Mumbai, 43.0 C):\n");
    insert(&head,"13:30","Mumbai",43.0);
    display(head);

    printf("\nSearching for 12:00:\n");
    search(head,"12:00");

    printf("\nUpdating 11:00 temperature:\n");
    update(head,"11:00",39.5);
```

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```c
    printf("\nUpdating 11:00 temperature:\n");
    update(head,"11:00",39.5);
    display(head);

    printf("\nDeleting 10:00 record:\n");
    delete(&head,"10:00");
    display(head);

    return 0;
}
```